

Matrix functions and stability

A uniform disk bound for wave-kernel Padé approximants

Difficulty: hard **Importance:** interesting to specialist

Status: open; original conjecture plus a search for subsequent resolution

Problem statement

Let

$$f(z) = \sum_{j=0}^{\infty} \frac{z^j}{(2j)!} = \cosh \sqrt{z},$$

where the series defines the entire function without a square-root branch choice. For each integer $m \geq 1$, let $r_m = P_m/Q_m$ be its diagonal Padé approximant at zero: $\deg P_m, \deg Q_m \leq m$, $Q_m(0) = 1$, and $Q_m(z)f(z) - P_m(z) = O(z^{2m+1})$. Is it true that the reduced rational function r_m has no pole in $\{z \in \mathbb{C} : |z| \leq 3\}$ and

$$|1 - r_m(z)| \leq 2 \quad (|z| \leq 3)$$

for every m ?

Reference and status evidence

Nadukandi and Higham, [Computing the Wave-Kernel Matrix Functions](#), SIAM J. Scientific Computing 40(6) (2018), DOI, §4.2, Conjecture 4.6, manuscript p. 12. Lemma 4.5 establishes the finite range $m \leq 20$; §4.3 explains its role in backward-error analysis. Searches for the paper title with “conjecture” and for “Conjecture 4.6” with “cosh” and “Padé” located no general proof or counterexample. The status evidence is therefore weaker than a recent paper explicitly reaffirming the conjecture.